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المدرسة: ليلى عبابنة

BTC 421 LAB WORK

1- Stock solution for *in vitro* culture media preparation

2- Media preparation:

A- MS hormone free medium
B- Shooting medium (MS+0.5 mg/L GA3+ 2 mg/L BA+0.3 mg/L IBA)
C- Rooting medium (MS+0.5 mg/L GA3+ 2 mg/L IBA)
D- Cailus medium (MS+0.5 mg/L GA3+ 0.5 mg/L 2.4.D)
(MS+0.5 mg/L GA3+ 2 mg/L BA+0.5 mg/L IBA)

E- Water Agar medium

3- Chickpea seed surface sterilization and inoculation on water agar medium

4- Chickpea micropropagation using nodal fragmentation

A- Multishoot induction (proliferation)
B- Root induction

5- Chickpea callogenesis using different explants on callus inducing medium

6- Chickpea protoplast isolation and culturing

7- Transformation of carrot slices with *Agrobacterium tramifaciens*

8- DNA extraction from plant tissues (chickpea)

9- Alkalin method for Ti-plasmid isolation

10- PCR technique as a tool for plant pathogen detection and identification

TABLE 3.6

Stock solutions for Murashige and Skoog's medium (MS)[a]

Constituents	Amount ($mg\ l^{-1}$)
Stock solution I	
NH_4NO_3	33 000
KNO_3	38 000
$CaCl_2.2\ H_2O$	~~8800~~ 6645
$MgSO_4.7\ H_2O$	7400
KH_2PO_4	3400
Stock solution II	
KI	166
H_3BO_3	1240
$MnSO_4.4\ H_2O$	~~4460~~ → 3538
$ZnSO_4.7\ H_2O$	1720
$Na_2MoO_4.2\ H_2O$	50
$CuSO_4.5\ H_2O$	5
$CoCl_2.6\ H_2O$	5
Stock solution III[b]	
$FeSO_4.7\ H_2O$	5560
$Na_2.EDTA.2\ H_2O$	7460
Stock solution IV	
Inositol	20 000
Nicotinic acid	100
Pyridoxine HCl	100
Thiamine HCl	100
Glycine	400

[a] To prepare 1 l of medium take 50 ml of stock I, 5 ml of stock II, 5 ml of stock III, and 5 ml of stock IV.

[b] Dissolve $FeSO_4.7\ H_2O$ and $Na_2.EDTA.2\ H_2O$ separately in 450 ml distilled water by heating and constant stirring. Mix the two solutions, adjust the pH to 5.5, and add distilled water to make up the final volume to 1 l.

it is better to add vitamins and auxins after autoclaving. If there is a special reason to do so the substances may be sterilized by filtering their solutions (adjusted to the desired pH) through microfilters[2] of pore size 0.22—0.45 μm (see Sect. 2.3.2).

(c) The final volume of the medium is made up with distilled water.

(d) After mixing well, the pH of the medium is adjusted using 0.1 N NaOH and 0.1 N HCl.

(e) The medium is poured into the desired culture vessels. About 15 ml

[2] Filter assemblies of various sizes and filter membranes of various porosity are manufactured by Millipore Intertech Inc., P.O. Box 255, Bedford, MA 01730, U.S.A.

issue culture media[a]

	Media (amounts in mg l^{-1})[b]						
	White's[c]	Heller's[d]	MS[e]	ER[f]	B_5[g]	Nitsch's[h]	NT[i]
	—	—	1650	1200	—	720	825
	80	—	1900	1900	2527.5	950	950
	—	75	440	440	150	—	220
$\cdot H_2O$	—	—	—	—	—	166	—
$O_4.7\ H_2O$	750	250	370	370	246.5	185	1233
H_2PO_4	—	—	170	340	—	68	680
$H_4)_2SO_4$	—	—	—	—	134	—	—
$(NO_3)_2.4\ H_2O$	300	—	—	—	—	—	—
NO_3	—	600	—	—	—	—	—
$_2SO_4$	200	—	—	—	—	—	—
$H_2PO_4.H_2O$	19	125	—	—	150	—	—
Cl	65	750	—	—	—	—	—
	0.75	0.01	0.83	—	0.75	—	0.83
BO_3	1.5	1	6.2	0.63	3	10	6.2
$nSO_4.4\ H_2O$	5	0.1	22.3	2.23	—	25	22.3
$nSO_4.H_2O$	—	—	—	—	10	—	—
$nSO_4.7\ H_2O$	3	1	8.6	—	2	10	—
$nSO_4.4\ H_2O$	—	—	—	—	—	—	8.6
$n.Na_2.EDTA$	—	—	—	15	—	—	—
$a_2MoO_4.2\ H_2O$	—	—	0.25	0.025	0.25	0.25	0.25
oO_3	0.001	—	—	—	—	—	—
$uSO_4.5\ H_2O$	0.01	0.03	0.025	0.0025	0.025	0.025	0.025
$oCl_2.6\ H_2O$	—	—	0.025	0.0025	0.025	—	—
$CoSO_4.7\ H_2O$	—	—	—	—	—	—	0.03
$AlCl_3$	—	0.03	—	—	—	—	—
$NiCl_2.6\ H_2O$	—	0.03	—	—	—	—	—
$FeCl_3.6\ H_2O$	—	1	—	—	—	—	—
$Fe_2(SO_4)_3$	2.5	—	—	—	—	—	—
$FeSO_4.7\ H_2O$	—	—	27.8	27.8	—	27.8	27.8
$Na_2.EDTA.2\ H_2O$	—	—	37.3	37.3	—	37.3	37.3
Sequestrene 330Fe	—	—	—	—	28	—	—
anic							
ositol	—	—	100	—	100	100	100
icotinic acid	0.05	—	0.5	0.5	1	5	—
yridoxine HCl	0.01	—	0.5	0.5	1	0.5	—
hiamine HCl	0.01	—	0.1	0.5	10	0.5	1
lycine	3	—	2	2	—	2	—
olic acid	—	—	—	—	—	0.5	—
otin	—	—	—	—	—	0.05	—
icrose	2%	—	3%	4%	2%	2%	1%
-Mannitol	—	—	—	—	—	—	12.7%

wth regulators and complex nutrient mixtures described by various authors are not included here. The compositions of several a recommended for specific tissue and organ are given in relevant chapters.
centrations of mannitol and sucrose are expressed in percentage.
te (1963).
ler (1953).
ashige and Skoog (1962).
sson (1965).
nborg et al. (1968).
ch (1969).
ata and Takebe (1971).

2

TISSUE CULTURE MEDIA

Plants grown in nature get their nourishment from nutrients available in soil and also synthesised by the plant tissues themselves. Plants when grown *in vitro*, solely depend on the nutrients contained in the medium. Nutritional requirement of different species may vary, therefore,a wide range of culture media differing in their elemental compositions have been formulated for culture of plant cells/ tissue (Table 2.1).

Tissue cultures can be maintained either in liquid medium or semi-solid medium. In the former, plant material is immersed in the medium either partially or completely while in the latter plant material is placed on the surface of the medium.

Suspension cell cultures are intiated by transferring preferably friable callus to liquid medium containing organic and inorganic nutrients and growth hormones. Suspension cultures are regularly aerated either by bubbling sterile air or gentle agitation on an orbital shaker at between 90 and 150 rpm, with an orbital diameter of 1-2 cm. The culture of plant tissues in an agitated liquid medium facilitates gaseous exchange, removes any polarity of the tissue due to gravity and eliminates nutrient gradients within the medium and at the surface of the cells.

COMPOSITION

Cells, tissues and organs can only be cultured *in vitro* when all the required nutrients are available in the medium. A nutrient medium should contain inorganic nutrients, organic nutrients and growth hormones.

1. INORGANIC NUTRIENTS

A number of inorganic nutrients are required for normal growth of th. Besides, carbon, hydrogen and oxygen, other 12 elements essential for growth include nitrogen, calcium, potassium, phosphorus, sulphur, magnesium, iron, manganese, copper, boron, zinc and molybdenum. Out of these, former six elements are macronutrients as these are required in comparatively larger quantities (concentrations more than 0.05 mmol l^{-1}), while the latter six are classed as micronutrients as these are required in smaller quantities (less than 0.05 mmol l^{-1}). In White's medium iron is added in the form of Ferric sulphate ($Fe(SO_4)_3$) but in most of the other tissue culture media iron is supplied in the form of Fe. EDTA. Nitrogen is furnished in the medium in the forms of nitrate (oxidised) and ammonia (reduced).

Deficiencies of the macroelements N,P,S,K, Mg and Ca are more easily manifested in tissue culture when cells are cultured in liquid rather than on agar media since impurities present in the agar are considerable. Of all the mineral nutrients, the form of nitrogen is responsible for the most pronounced effects on growth and differentiation of cultured tissues.

2. ORGANIC NUTRIENTS

Cultured plants accomplish better growth when medium is supplemented with organic nutrients such as amino acids and vitamins. The most commonly used vitamin is thiamine (vitamin B_1). Other vitamins which improve the growth of cultured plants include nicotinic acid, calcium pantothenate and pyridoxine. Besides these, biotin, folic acid and aminobenzoic acid are added in various concentrations in some of the media.

As a carbon source, sucrose is the most preferred carbohydrate. Glucose, fructose, maltose, galactose, mannose and lactose are the other favourable sugars.

In order to promote growth of callus, some complex semisynthetic substances are added in the medium. These include yeast extract (YE), coconut milk (CM), casein hydrolysate (CH), tomato juice (TJ) and malt extract (ME).

3. GROWTH HORMONES

In addition to inorganic nutrients, vitamins and the carbon source, basal medium is supplemented with growth hormones such as auxins, cytokinins and gibberellins.

4

TABLE 2.1 Constituents of some common plant tissue culture media (mg litre⁻¹).

Constituents		WM	MS	NM	NT
Inorganic compounds					
Potassium chloride	KCl	65	-	-	-
Potassium nitrate	KNO_3	80	1900	950	950
Ammonium nitrate	NH_4NO_3	-	1650	720	825
Calcium chloride	$CaCl_2.2H_2O$	-	440	166	220
Calcium nitrate	$Ca(NO_3)_2$	300	-	-	-
Magnesium sulphate	$MgSO_4.7H_2O$	720	370	185	1233
Sodium sulphate	$Na_2SO_4.10H_2O$	200	-	-	-
Potassium dihydrogen phosphate	KH_2PO_4	-	170	68	680
Sodium dihydrogen phosphate	$NaH_2PO_4.7H_2O$	165	-	-	1233
Manganese sulphate	$MnSO_4.4H_2O$	3	22.3	25	22.3
Boric acid	H_3BO_3	1.5	6.2	10	6.2
Zinc sulphate	$ZnSO_4.7H_2O$	3	8.6	10	8.6
Potassium iodide	KI	0.75	0.83	-	0.83
Sodium molybdate	$Na_2MoO_4.2H_2O$	-	0.25	0.25	0.25
Copper sulphate	$CuSO_4.5H_2O$	0.01	0.025	0.025	0.025
Cobalt chloride	$CoCl_2.6H_2O$	-	0.025	-	-
Cobalt sulphate	$CoSO_4.7H_2O$	-	-	-	0.03
Disodium ethylene diamine tetra-acetate	$Na_2EDTA.2H_2O$	-	37.3	37.3	37.3
Ferrous sulphate	$FeSO_4.7H_2O$	-	27.8	27.8	27.8
Ferric sulphate	$Fe(SO_4)_3$	2.5	-	-	-

(Contd.)

Constituents	WM	MS	NM	NT
Organic compounds				
Inositol	-	100	100	100
Nicotinic acid	0.05	0.5	5	-
Pyridoxine HCl	0.0	10.5	0.5	-
Thiamine HCl	0.01	0.1	0.5	1
Glycine	3	2	2	-
Ca-D-pantothenate	1	-	-	-
Sucrose	20,000	20,000	20,000	20,000
Mannitol	-	-	-	13%w/v
Growth regulators				
IAA	-	1.30	-	-
NAA	-	-	-	3
Kinetin	-	0.4-10	-	-
6 benzyl amino purine	-	-	-	1

WM - White's medium (1954)
MS - Murashige and Skoog medium (1962)
NM - Nitsch' medium (Nitsch, 1969)
NTM - Nagata and Takebe's medium (Nagata and Takabe, 1972)

Auxins promote cell division and root differentiation. Most commonly used auxins are: IAA (indole acetic acid), IBA (indole-3-butyric acid), NAA (naphthalene acetic acid) and 2, 4-D (dichlorophenoxy acetic acid). Cytokinins are responsible for cell division and shoot differentiation. BAP (benzyl-amino purine), and 2-isopentanyl adenine are most widely used cytokinins. Among gibberellins, GA_3 is commonly used.

4. AGAR

Agar (a polysaccharide obtained from seaweeds) is used to solidify the medium. Agar is commonly used at a concentration of 0.8-1% (w/v). Use of higher concentration of agar makes the medium hard. This prevents the diffusion of nutrients into the tissues.

5. pH

The pH of the medium is adjusted between 5 to 5.8 by adding (O.1N) NaOH or HCl. Usually a pH higher than 6 results in a fairly hard medium whereas a pH below 5 does not allow satisfactory solidification of the medium.

MEDIA PREPARATION

The simplest method of preparing culture media is to use commercially available powdered media. These media contain all the requisite nutrients. However, agar, sugar and other supplements are added in the finally prepared media.

Another method of preparing media is to prepare concentrated stock solutions. For the preparation of stock solutions, distilled water and chemicals of high purity are used. Stock solutions varying in strength are prepared of different media components viz., inorganic salts, vitamins and other organic compounds and growth regulators. All the stock solutions are stored either in glass or plastic containers and stored in refrigerator. Stock solution of iron is stored in amber-coloured bottles. Substances which are not stable in frozen state must be freshly prepared each time the medium is required and added to the final mixture of stock solutions.

For the preparation of medium, the following sequencial steps are followed:

1. Appropriate quantities of agar agar and sucrose are dissolved in distilled water.
2. Required quantities of stock solutions, growth hormones and other supplements are added.
3. More distilled water is added to make the final volume of the medium.
4. pH of the medium is adjusted between 5 - 5.8 by adding 0.1N NaOH or 0.1N HCl.
5. The medium is poured in culture tubes, flasks or any other containers.
6. Culture vessels are plugged with non-absorbent cotton wool wrapped in cheese cloth. This will allow free gaseous exchange but inhibit microbial contamination.
7. Culture vessels are autoclaved at 120°C (1.06 kg/cm^2) for 15-20 minutes.
8. Culture medium is allowed to cool at room temperature and used or stored at 4°C.

Media Selection

As there is no single medium suitable for culturing all cell, tissue and org it is necessary that the most suitable medium be determined to accommoda specific requirements of the plant material in question. It is always better to begin with well known basal medium (Table 2.1). To select a suitable medium for an untested system, broad spectrum experiments may be setup as suggested by De Fossard *et al.* (1974). In this experiment all the components of the medium are divided into four major groups: (a) minerals, (b) auxins, (c) cytokinins and (d) organic nutrients. For each major group, three concentrations (low, medium, high) are selected. Various combinations of four groups and three concentrations result in most suitable media for a particular species.

STERILIZATION

The media used for plant tissue cultures also provide suitable environment for the growth of microorganisms like bacteria and fungi. These organisms grow faster than the cultured tissues and produce toxic metabolites which inhibit the growth of explant or even kill it. Therefore, it is absolutely essential to provide and maintain a completely aseptic environment inside the culture vessels.

Sterilization Room

The dimensions of sterilization room should be 6x6x4.5 sq.m. The room should be equipped with an autoclave, either of horizontal or vertical type. Usually former type of autoclaves are preferred. The room should have a good ventilation.

Sources of Contamination

There are a number of sources through which medium may get contaminated. These include medium itself, explant, culture vessels and instruments, environment of culture room and place of transfer.

(a) **Medium:** The microbes may be present in the medium right from the very beginning. To eliminate them, vessels containing the medium are plugged and autoclaved at 120°C for 15-20 minutes. Sterilization is done in steel autoclaves or in pressure cookers if former is not available.

Some growth substances and vitamins are thermo-labile. Thus they can not be autoclaved. Such compounds are sterilized by passing through membrane filters of 0.22 μm pore size. Small volumes can be sterilized using syringe holder but large volumes (above 500 ml) require vaccum filtration.

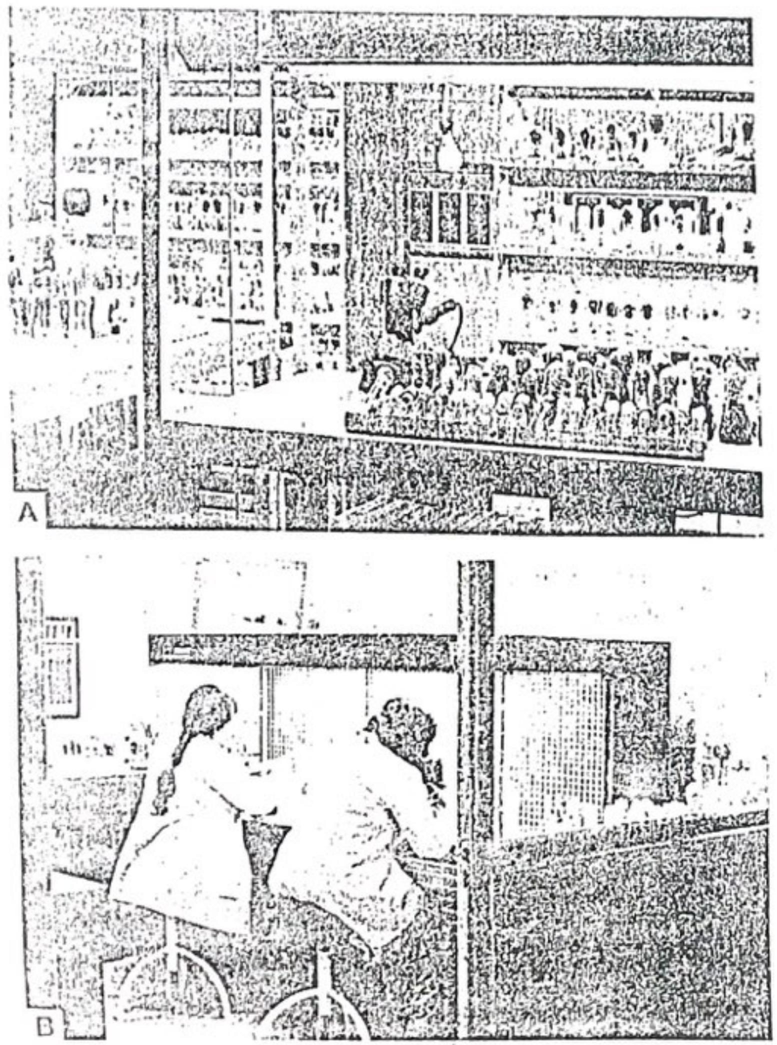

Fig.2. A. A view of one of the culture rooms of the Tissue Culture Laboratory, National Botanical Research Institute, Lucknow, B. Inoculation work being performed in a laminar flow cabinet.

(b) **Culture vessels and instruments:** Glasswares and metal instruments are sterilized by placing them in an oven at 160-180°C for 3 - 4 hours. The instruments such as scalpel, forceps, needles etc., dipped in 95% ethyl alcohol, are flammed before use.

(c) **Plant material:** The plant tissue which is to be cultured must be surface sterilized before placing it on the medium. This is essential to avoid microbial contamination. Usually calcium or sodium hypochlorite solutions are used. In cases where plant material is very hard such as seeds, treatment with disinfectant may be done directly. On the other hand, while culturing young embryos or ovules, surface sterilization of protecting tissues (e.g. ovules in case of embryos and ovary in case of ovules) are done to avoid damage to the explant.

(d) **Transfer area:** Aseptic manipulations are conducted in a closed transfer area fitted with gloves and UV lamps. Ultra violet light sterilizes the interior portion of the inoculation chamber and also eliminates atmospheric contamination. These days, in most of the tissue culture laboratories, laminar air flow cabinets are used. Air passes through ultrafilters which flows in the working area at a constant speed. These filters not only eliminate dust and other particles but also fungal and bacterial spores. Thus an aseptic environment is maintained at the time of tissue inoculation.

Experiment 2

STERILIZATION OF PLANT MATERIAL

Requirements

Explant plant material, Calcium/Sodium hypochlorite, 0.1% mercuric chloride, ethanol (70% v/v), screw cap bottles, or sterile flask, spatula sterilized forceps, scalpels, sterilized distilled water.

Procedure

1. Cut small pieces of plant material (Seedlings, bud stem, root, leaf) using scalpel.
2. Wash explants under running water (root, etc.)
3. Rinse explants in 70% ethanol for a few seconds and expose the material under the sterile hood for evaporation of ethanol.
4. Transfer the explant material into a vial/sterile flask containing 5% Sodium/Calcium hypochlorite solution or 0.1% $HgCl_2$ under aseptic conditions.
5. [illegible]ake the vial/flask slowly, so that all the surface of the [illegible]ants comes in contact with the solution.Keep the [illegible]aterial for 5-30 min.
6. Decant the $NaOCl/Ca(OCl)_2/HgCl_2$ solution.
7. Rinse the explant material with sterile distilled water 4-5 times.

Note —

- In this procedure only surface sterilization of the explant material can be performed.
- If the plant material is seed, skip the first 3 steps.
- If the plant material is obtained from sterilized, aseptic conditions then there is no need to sterilize the material again.
- Seed-borne fungal infections cannot be controlled by this method. Seedling explants must be sterilized if the infection persists after germination of sterile seeds.
- The timing(in step 5) of sterilization varies with explant type and also with the sterilizing agent.
- 0.1% $HgCl_2$ - Dissolve 100 mg of $HgCl_2$ in 100 ml dd H_2O

LAB # 3

chick pea Seed Sterilization

1 - runing tap water 5 min

2 - Soap 5 min

3 - Remove extra soap

4 - Ethanol 70% 30 sec

5 - Washing with D.W

6 - Clorex 1 - 4% (Sodum Hypoclorate) 15 min

7 - Washing with autoclave D.W 20 min X3

under laminar flow

14

LAB # 4

8

MICROPROPAGATION

In nature asexual reproduction takes place either by vegetative means (regeneration of new plants from vegetative parts) or by apomixis (seed development without meiosis and fertilization). The vegetative reproduction produces genetically identical plants and is widely used for the propagation and multiplication of horticulturally important plants. The multiplication of genetically identical copies of a cultivar by asexual means is called **clonal propagation** and a genetically uniform assemblage of individuals derived originally from a single individual by asexual propagation constitutes a **clone**. The word clone (Greek clone = twig, spray or a slip, like those broken off as propagules for multiplication) was first used by Webber (1903) to apply to cultivated plants that were propagated vegetatively. *In vitro* clonal propagation is called **micropropagation**.

Micropropagation has a number of advantages over conventional methods of clonal propagation such as (1) a large number of plants can be produced from a single explant within a short period and space; (2) *in vitro* cultured plants are disease free and (3) micropropagation may be carried out throughout the year irrespective of seasonal variation.

MICROPROPAGATION TECHNIQUES

The process of micropropagation involves four distinct stages:

1. Selection of suitable explants, their sterilization and transfer to a nutrient medium,

2. proliferation of shoots on multiplication medium,
3. transfer of shoots to a rooting medium and
4. transplantation of plants in soil (Figs. 8.1, 8.2).

SHOOT MULTIPLICATION

In vitro multiplication of shoots involves three main approaches : (1) multiplication through callus culture, (2) multiplication by adventitious shoots and (3) multiplication by apical and axillary shoots.

1. Multiplication through Callus Culture

A large number of plantlets can be obtained from callus either through shoot-root formation or somatic embryogenesis. Mass production of callus followed by shoot regeneration would be the ideal method for large scale propagation of desired plants but there are two serious draw backs : (1) due to repeated subcultures the capacity of mass calli : to regenerate shoots is diminished or even lost, (2) degree of aneuploidy, polyploidy and development of genetically aberrant cells progressively increases resulting in the regeneration of plants that differ from the parent type. Therefore, multiplication of shoots through callusing is less preferred method.

2. Multiplication by Adventitious Shoots

Adventitious shoots are stem and leaf structures which develop naturally on plants at places other than normal leaf axil regions. These structures include stems, tubers, bulbs, corms and rhizomes. In many horticultural plants, vegetative propagation through adventitious bud formation is commercial practice. For example, in nature shoots develop on leaves of *Begonia* and some other ornamental plants. In culture also similar type of adventitious shoot formation can be induced by using appropriate combination of growth regulators in media. Adventitious buds can also be induced on the leaf and stem cuttings of even those species which are normally not propagated vegetatively e.g. flax (*Linum usitatissimum*), *Brassica* species.

Adventitious buds may be formed from a piece of shoot apex. In grapes, from an excised shoot tip more than 8000 plants can be obtained within 3-4 months. *Pinguicula* is usually propagated by leaf pieces and in conventional methods only one plant is produced from each leaf but in cultures 15 fold increase in propagation has been achieved.

African violet (*Santpaulia ionantha*) is conventionally propagated from leaf cuttings which give rise to one or more shoots developing from the base of each petiole. *In vitro* culture of petiole can yield upto 20,000 plantlets from each petiole. Similarly large scale propagation of *Petunia hybrida* and *Kalan-*
[illegible] small leaf or stem pieces.

Development of adventitious shoot directly from excised organs is preferred for clonal propagation as compared to callus formation method because in former diploid individuals are formed but in latter often cytologically abnormal plants are produced. However, in varieties which are genetic chimeras, multiplication through adventitious bud formation may lead to splitting the chimeras to pure type plants.

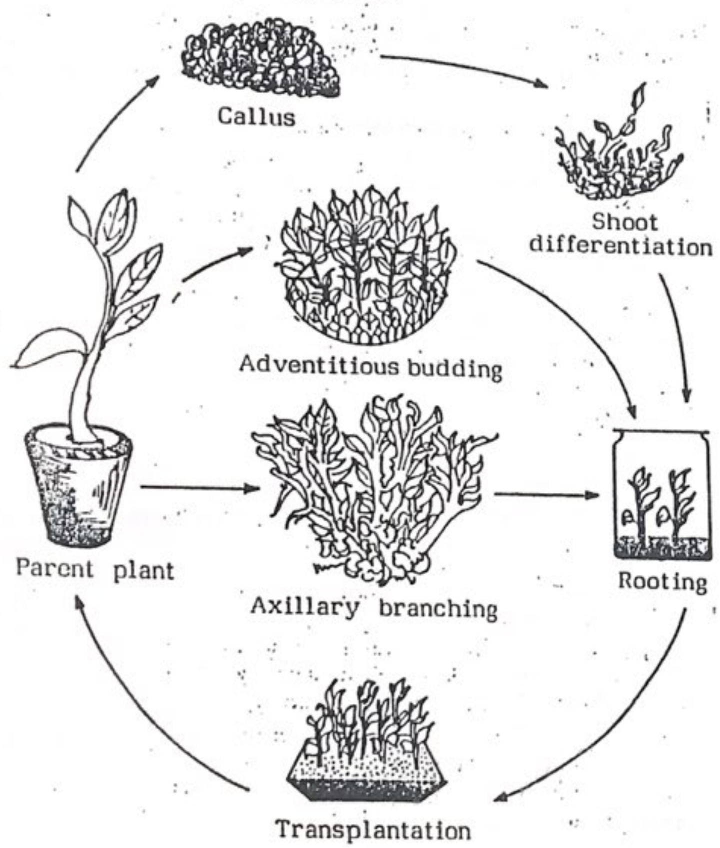

Fig.8.1. Diagrammatic summary of steps involved in aseptic multiplication of plants. Shoot multiplication is achieved through axillary branching, adventitious budding from pieces of plant parts or callusing.

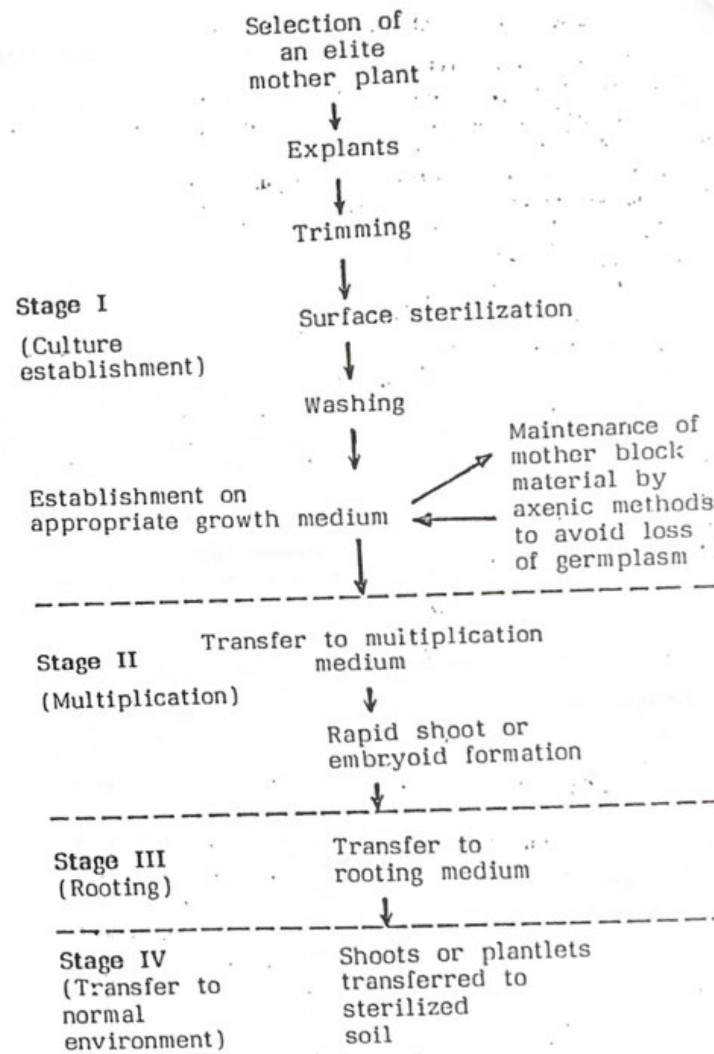

Fig.8.2. Stages of micropropagation.

16

3. Multiplication by Apical and Axillary Shoots

Apical shoots are those that occupy the growing tip of shoots whereas axillary shoots are those that develop from their normal positions on the plant in the axil of each leaf. Every bud has got the potentiality to develop into a shoot. Apical dominance plays an important role in the development of axillary buds and is governed by interplay of growth regulators. In the species where apical dominance is very strong, removal or injury of terminal bud is necessary for the growth of axillary buds.

To initiate culture shoot tips are placed on the medium containing low levels of cytokinins (0.05-.5 mgl^{-1} BAP) and auxin (0.01-0.1 mgl^{-1} IBA), the level of cytokinin being progressively raised at each subculture until desirable rate of proliferation is achieved.

In cultures the rate of shoot multiplication can be enhanced by culturing shoot tips on the medium containing cytokinin. In such cases the cultured explant transforms into mass of branches. By further subculturing shoot multiplication cycle may be repeated and cultures may be maintained round the year. Proliferating shoots of many species have been maintained upto 10-15 years.

ROOTING OF *IN VITRO* SHOOTS

Adventitious and axillary shoots formed in cultures do not possess roots. Therefore, they must be transferred to a root initiating medium which differs from shoot multiplication medium in its growth hormone composition. The basal cut end of the shoot is treated with root initiating powder and planted in the potting mix.

Initiation of rooting is often the most labour-intensive stage of *in vitro* propagation because the shoots have to be handled singly rather than in proliferating clusters. All cytokinins inhibit rooting. The use of kinetin in place of BAP in final stage of multiplication improves rooting. Rooting is also improved by lowering the concentration of macro salts to a half or less, and the concentration of sucrose from 2 or 3% to 0.5 or 1%.

TRANSPLANTATION

The plants are carefully removed from the medium and transferred to the soil. High humidity (90 - 100%) must be maintained for 10 - 15 days. This may be done by covering the pots with polythene covers having holes for aeration.

The best time for transplanting is when the roots are beginning to emerge out and the developing leaves are becoming photosynthetically self-supporting.

MICROPROPAGATION OF ELITE PLANTS

Tissue culture technique is becoming popular for propagation of a large number of economically important plants. (Fig. 8.3). In orchids, tissue culture is the only commercially feasible method for large scale propagation.

Association of a fungus is a must for orchid seed germination *in vivo* and also the plants obtained through seeds take 4-5 years time to reach flowering stage. In order to overcome these problems, *in vitro* propagation of orchids is the best possible method. Morel (1960) cultured shoot tips of orchid *Cymbidium* and obtained adventitious protocorms. These protocorms by further subculturing may produce four million plants in a year.

Successful micropropagation has been achieved in a number of elite trees such as poplar, willows, *Eucalyptus, Santalum, Phoenix* and *Tectona*. A serious problem associated with the culture of tree species is the browning of the culture medium. This occurs due to the oxidation of phenolic substances which leach out from the cut surfaces of the explants. This problem has been overcome by culturing the explant in liquid medium for a few days or treating the explant with antioxidant solutions before culture.

Many commercial companies are producing millions of plantlets of ornamentals, woody, vegetable and crop species using micropropagation techniques (Table 8.1).

Table 8.1. Some genera micropropagated in *In vitro*.

Vegetable and crop plants	Ornamental plants	Woody plants
Allium	*Anthurium*	*Betula*
Arachis	*Bromeliads*	*Coffea*
Asparagus	*Chrysanthemum*	*Eucalyptus*
Beta	*Cymbidium*	*Hevea*
Brassica	*Dianthus*	*Malus*
Cynara	*Ferns*	*Phoenix*
Fragaria	*Gerbera*	*Pinus*
Glycine	*Gladiolus*	*Populus*
Linum	*Iris*	*Prunus*
Phaseolus	*Narcissus*	*Salix*
Solanum	*Orchids*	*Santalum*
Vigna	*Phlox*	*Tectona*
Zea	*Tulipa*	*Vitis*

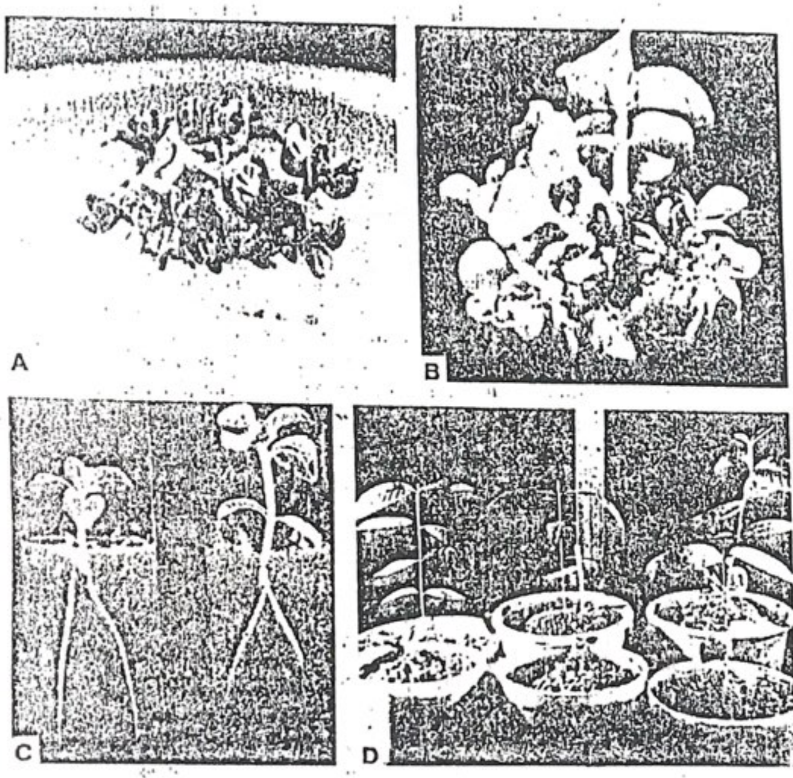

Fig.8.3. Micropropagation of guava. **A.** Multiple shoot formation from the nodal segments of guava on MS medium containing 1.0 mg l^{-1} BAP after 6 weeks. **B.** Same after 8 weeks of culture. **C.** Rooting of microshoots on medium containing 0.2 mgl^{-1} each of IBA and NAA with activated charcoal after 3 weeks. **D.** Different growth stages of regenerants after 3 to 12 weeks of transfer on soil.

FACTORS AFFECTING MICROPROPAGATION

1. Donor Plant and Explant

Physiological condition of the donor plant and explant greatly affect morphogenesis and rate of proliferation in micropropagation systems. Explants which contain large proportion of meristematic tissue are most suitable. The explants obtained from younger parts of the plants give better result than older ones. The behaviour of explants in culture is determined by the stage of the growth of parent plant and season of obtaining the explant.

Most plants can be readily cultured when explants from actively growing plants are used, but others may produce shoots only from plants that are dormant. The bulb scales of *Lilium speciosum* regenerated bulbils freely when taken from plants during spring and autumn periods of growth, but not when taken during summer or winter.

2. Media

The overall micropropagation performance of an explant is directly controlled by the composition of the medium. Higher levels of cytokinins may promote shoot formation but the growth of individual shoots is arrested. In tropical foliage plants, the concentration of phosphate present in the medium affects the size and number of shoots produced in shoot tip cultures. Likewise, the level of potassium in the medium influences the number of somatic embryos formed in wild carrot.

Activated charcoal at about 2% is often included in nutrient media, its action being to absorb aromatic compounds. Charcoal prevents browning in cultured tissues and stimulates embryogenesis and rooting.

3. Light

Light plays an important role in inducing morphogenesis in cultured tissue. The most suitable light intensity for micropropagation is 1000 lux. Low light intensities result in green and taller shoots while high light intensity induces better rooting.

More adventitious shoots were regenerated on *Allium cepa* tissues cultured in 16-hour day than 8-hour day. With shoot cultures of *Solanum tuberosum* the greatest rate of node formation occurred in continuous light, fewer nodes and weaker shoots being formed in 8-hour day.

4. Temperature

Tissue cultures are usually maintained at 25°C. However, some plant species require varied temperature range for better initiation of shoot and roots. In *Asparagus* shoot tip cultures grew best at 27°C whereas petiole segments of *Begonia* and *Cheimantha* cross produced greatest number of shoots at 18°C.

Tropical species require higher temperature, the optimum for date palm being 27°C and for *Monstera deliciosa* 30°C.

APPLICATIONS OF MICROPROPAGATION

1. *In vitro* propagation is a powerful and attractive tool for the rapid cloning of desirable plants. Newly selected cultivars can be bulked up rapidly to the level of several thousands in months rather than years. Additionally, in tissue cultures plant multiplication can continue throughout the year irrespective of the season.
2. Some heterozygous horticultural crops such as *Gerbera* and *Anthurium* that are usually propagated only by seeds can now be cloned rapidly from individuals selected for vigour and colour.
3. Majority of vegetatively propagated crops are susceptible to virus infection. The advantageous use of micropropagation here lies in the rapid production of disease free material. The technique of "meristem tip culture" combined with heat treatment of the mother plant, is now the established means of virus elimination from infected stock.
4. In some crops production of hybrid seeds is very expensive. Production cost of such seeds may be reduced by *in vitro* multiplication of selected parents. Broccoli, cabbage and cauliflower are examples of plants that can now be propagated *in vitro* and desired parents multiplied for the direct production of hybrid seed.
5. Micropropagation has great value as a potential system of germ plasm storage.
6. Tissue cultures can be used to minimise the growing space in commercial nurseries for the maintenance of stock plants.

9

ANTHER CULTURE AND PRODUCTION OF HAPLOIDS

Anther culture is a means of obtaining haploids from pollen grains. As these haploids contain a single set of chromosomes, they are of special interest to geneticists and plant breeders for developing new crop cultivars. Under appropriate cultural conditions, the microspores within the anthers of some species can be induced to undergo embryogenesis resulting in the formation of embryos from which plants can be regenerated (Fig. 9.1). Haploids have been obtained from anther cultures of a number of species but taxa belonging to families Solanaceae, Poaceae and Brassicaceae respond better in cultures. Pollen plants have also been successfully induced in several woody genera such as *Citrus, Vitis, Hevea* etc.

The first pollen-derived haploid callus was obtained in anther cultures of *Tradescantia reflexa* by Yamada *et al.* (1963) but the callus failed to undergo organogenesis. The induction of haploid embryos from pollen grains of *Datura innoxia* by Guha and Maheshwari (1964, 1966) paved the way for production of androgenic haploids. Bourgin and Nitsch (1967) were the first to obtain mature flowering haploid plants of pollen origin in *Nicotiana tabacum* and *N. sylvestris*. Haploid production through anther and pollen culture has been reported in more than 125 species (Table 9.1). Even in those species where *in vitro* androgenesis has been reported, often the frequency with which haploids are formed is very low.

19

LAB #5

4

CALLUS GROWTH PATTERNS

Callus may be defined as an unorganised mass of loosely arranged parenchymatous cells which develop from parent tissue due to proliferation of cells (Fig. 4.1).

CALLUS INITIATION

Callus initiation begins when segments (explants) of plant organs are cultured on solidified culture media. Callus cultures can be initiated from explants taken from any part of the plant such as root, stem, leaf, seed etc. Such cultures can be maintained for an indefinite period if they are transferred to fresh nutrient media at regular intervals. This is known as subculturing.

Explants from carrot root have been most widely used to demonstrate callus initiation. Gautheret (1959) observed that the callus formation by carrot root tissues is related to the region from which the explant has been removed. For example, explants of xylem and phloem adjascent to the cambium region show vigorous growth and produce enormous amount of callus.

A procedure for the initiation of callus culture from tobacco stem tissue is outlined in figure 4.2.

When explant is placed on the medium, cellular masses begin to appear around the cut edges of the explant. The initiation of callus formation is visible when the surface of the explant begins to glisten and the texture becomes rough. Repeated cell divisions cover the entire explant with unor-

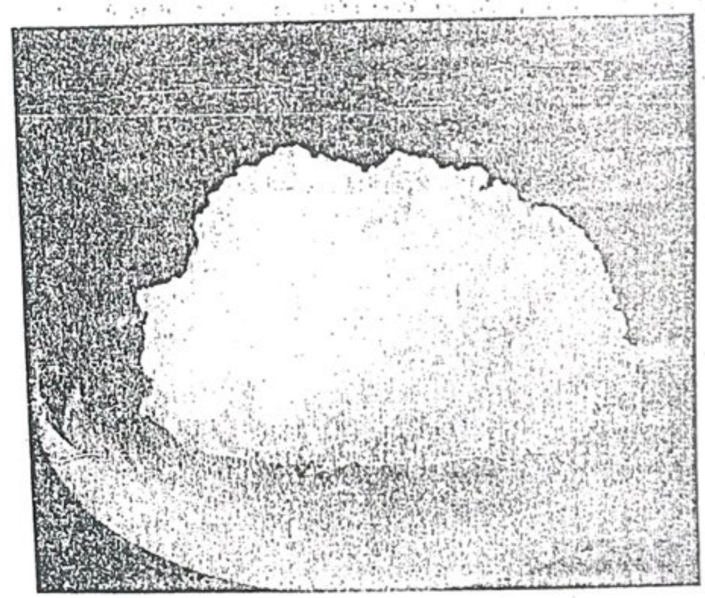

Fig.4.1. A typical friable callus-habituated stem callus of *Citrus grandis*.

21

ganised callus in 2-3 weeks (Fig. 4.3). The cell division increases the size of the explant. If the callus is allowed to grow on the same medium for a longer period, depletion of nutrients and desiccation of agar will occur which will affect the further growth of the callus. Besides this, developing callus may also produce metabolites which may have toxic effect. This may be overcome by subculturing the callus. The callus is divided into 2-4 equal segments and transferred onto fresh culture medium of the same composition. After several such subcultures, callus gets established and its growth gets stablized.

Stem segment
↓
Sterilization
↓
Rinsing in distilled water
↓ Cut
Stem discs
↓
Plating on agar solidified medium
↓
Callus

Fig.4.2. Initiation of callus.

Callus cultures may be initiated from any meristematic tissue but all callus cultures do not produce plants. In grasses, however, callus cultures derived from immature embryos show highest rate of plant regeneration.

Callus cultures which comprise unorganised mass of cells may be subcultured to produce more calli or may be induced to produce roots, shoots or embryoids which further get differentiated to form new plants.

In cereals (wheat, rice, maize and sorghum) seedling segments may be induced to produce callus. These cultures initially remain unorganised but subsequently develop roots. When callus growing on agar medium was

transferred to liquid medium of same composition, a large number of roots with lateral roots developed. But when this differentiating tissue was transferred to the agar medium, they reverted back to unorganised tissue and further developed as callus.

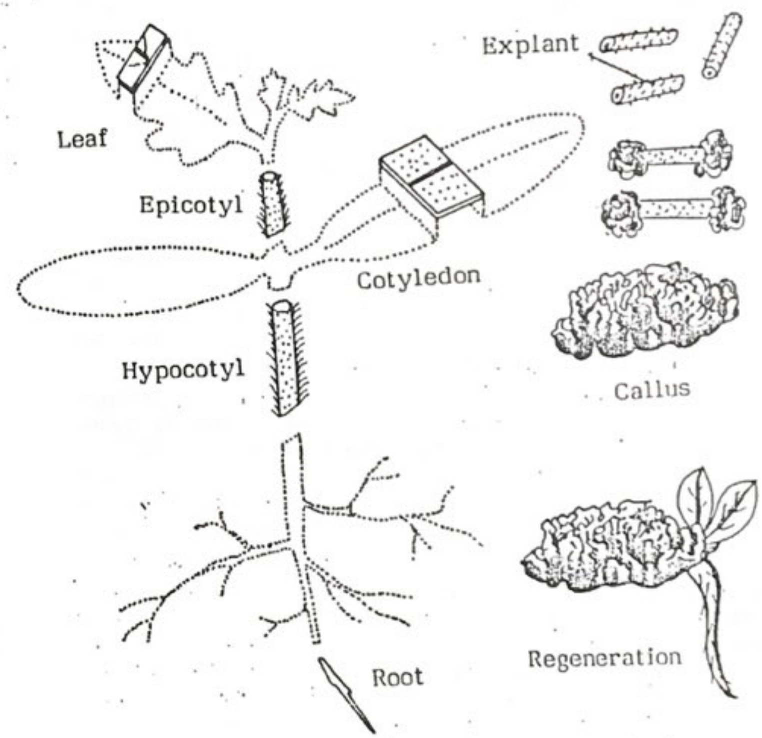

Fig.4.3. Locations of explant sources, callus initiation and regeneration.

2.

CALLUS GROWTH PATTERNS

The growth of callus is measured in terms of increase in fresh weight, dry weight or increase in cell number. A generalized growth pattern takes the form of a sigmoid curve (Fig. 4.4). Three distinct phases can be observed during growth of the callus : (a) the lag phase (b) cell division followed by linear phase and (c) stationary phase and regeneration. Although each stage is characterised by distinct structural and biochemical features but the entire growth cycle is in reality a continuum of physiological changes.

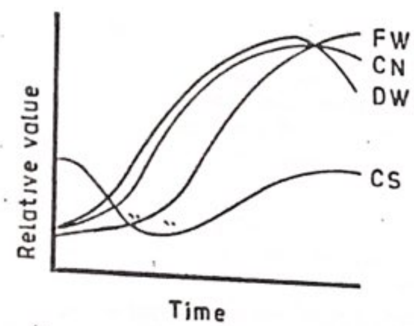

Fig.4.4. Features of a generalized growth pattern of cultured plant cells (CN= cell number, CS= mean cell size, DW= dry weight, FW= fresh weight)

The Lag Phase

This is a period of little or no cell division. When cultured cells are transferred to a fresh nutrient medium, a number of metabolic changes occur. These changes prepare the cells to undergo further division. The levels of NADPH, ATP, cell RNA content and energy charge increase. The rate of total protein synthesis also increases during the lag phase. This increase in protein synthesis is partly due to the efficient ribosomal translation capacity. In *Phaseolus* and *Glycine* subcultures result in rapid formation of polysomes and increase in the translatable levels of group of mRNAs. Presence of auxins 2, 4-D and NAA in the medium increase the mRNA.

During lag phase, only low levels of new compounds are synthesized. An enzyme phenylalanine ammonia-lyase is rapidly synthesised which enhances the activity during lag phase. However, this activity decreases as the culture completes lag phase. Accumulation of complex phenolic compounds do not occur during lag phase.

The Cell Division and Linear Phase

The lag phase is followed by a period of exponential cell division which is followed by a steady rate of division (the linear phase). Once cell division is induced it progresses rapidly. The cell division and expansion are usually confined to the peripheral region of the callus tissue. The number of cells increases which leads to sharp decrease in average cell size. Proliferation is mostly located either in sheet meristems or in nodules at the surface of the callus. When the cell division activity slowsdown, the meristematic activity begins in the deeper tissues of the callus. Chlorophyll formation decreases in rapidly dividing cells. In tobacco, linear phase cells contain poorly developed chloroplasts.

The growth rate of callus tissue can be enhanced by increasing the auxin concentration of the medium. However, the growth rate may be restricted by limiting the levels of essential nutrients in the medium.

The Stationary Phase and Regeneration

As the nutrient supply of the medium depletes, a gradual cessation of cell division occurs. There is sharp decline in the rate of respiration and in DNA synthesis as well as activities of enzymes associated with the pentose phosphate pathway.

Usually the stationary phase is associated with the initiation of structural organisation of the cells which increases the production of secondary metabolites such as alkaloids. Disorganised callus cultures accumulate lesser secondary metabolites. Lindsey and Yeoman (1983) observed that in many Solanaceous taxa at the end of growth phase, organisation and alkaloid accumulation increase together. This supports the view that cellular organisation provides favourable environment for the synthesis and accumulation of secondary metabolites.

The structural and biochemical differentiation result in the regeneration of different organs. Successive subcultures, however, affect the ability of regeneration. In *Hevea brasiliensis* new cultures develop roots spontaneously but this ability of cultures is lost during serial subcultures. Similiarly, single cell clones of tabacco gradually loses the ability to initiate roots and shoots. The potential for secondary metabolite production declines with serial propagation. In cell cultures of *Atropa belladona* accumulation of alkaloid which is dependent upon the formation of root, decreases with subculturing. Prolonged cultures are capable of synthesizing alkaloids. In *Datura innoxia* even after 93 weeks in culture, the callus was capable of synthesizing tropane alkaloids and chlorophyll.

In cultures of immature embroys of *Panicum*, the scutellum produced a compact callus that produced several cup-shaped structures which subsequently differentiated into embryoids. These embryoids germinated when auxin concentration was either reduced in the medium or omitted completely. On the other hand, callus obtained from mature embryos gave rise to shoot buds followed by differentiation and advetitious buds. For induction of embryogenic callus, the developmental stage of the inflorescences where floral primordia have just formed showed best response.

NUTRITIONAL REQUIREMENT

A nutrient medium is needed for the initiation and maintenance of callus cultures. Usually basal medium which consists of a mixture of mineral salts, vitamins, sucrose and phytohormones is used for initiating the callus. The basal medium is supplemented with coconut milk, yeast and malt extracts for additional nutritional requirement of the callus. In forage grasses, supplementation of MS (Murashige and Skoog, 1962) medium with only 2,4-D (2,4 Dichlorophenoxy-acetic-acid) is sufficient to induce callus formations. The continued proliferation and growth of the callus in subsequent subcultures need additional growth hormones or reduced level of 2, 4-D.

For callus induction and proliferation requirement of different growth hormones varies viz., addition of 2, 4-D is essential for callus induction and proliferation. It is not required at the time of organogenesis and plantlet formation. Usually, a high concentration of auxin and low concentration of kinetin promote callus induction and proliferation.

Tissues derived from the green photosynthetic organs may grow in cultures without addition of exogenously supplied sugar. However, for continued growth of callus, presence of exogenously supplied sugar becomes indispensable. Israel and Steward (1966) reported that callus cell in cultures are deficient in some enzymes and cofactors which are essential for synthesis of chlorophyll or proper functioning of chloroplast.

VARIATION IN CALLUS CULTURES

The genetic stability of callus cultures is one of the most important factors that determines the successful utilization of cell cultures for clonal propagation. It has been observed that spontaneous changes occur in the genetic makeup of the prolonged callus cultures. These spontaneous changes may include cytoplasmic structures such as mitochondria and plastids or chromosomes. The cytoplasmic changes may occur either through assymetric cell division or mutation. The chromosomal changes may involve increase in chromosome number, chromosomal breakage, gain, loss or reunion. In callus

subcultures fusion of nuclei may takes place which results in the increase of the chromosome number. The rate of these changes vary with the composition of the culture medium, the age of the culture and the species from which explant has been obtained. Some subculture calli do not show any change in the ploidy level even if they are cultured for a period of two years but there are instances where calli show high level of polyploidy within a few months of culturing the explant.Cell suspension prepared from single cell cultures derived from 5 varieties of sugarcane were shown to have variation in chromosome number and karyotype after 6 years in culture.Variation observed among plants regenerated from gametic, somatic and protoplast cultures have been termed respectively as gametoclonal, somatoclonal and protoclonal variation. Somaclonal variation has been attributed to variation already present in the plant material itself and/or to the changes occur during the tissue culture cycle. The latter involves establishment of a more or less dedifferentiated cell mass (e.g. callus), proliferation of a number of cell generations, followed by regeneration of plants.

In *Datura innoxia* plants have been regenerated successfully from aneuploid and polyploid callus without affecting the alkaloid pattern and content of the plants. In tobacco, the plantlets regenerated from the pith of autotetraploid callus accumulated slightly higher level of alkaloid nicotine as compared to the diploid plants of the same species. In callus cultures, therefore, at every stage of cell development, structural and biochemical heterogeneity occur.

Somaclonal Variation

Larkin and Scowcroft (1981) coined the term somaclonal variation to include all those variations which occur in plants regenerated from somatic cultures. Somaclonal variation seems not to be species or organ specific and variation among somaclones have been observed for a wide array of characters. Chaturvedi and Mitra (1975) described two subclones of *Citrus grandis* callus which differed in their morphogenetic pattern. Under identical culture conditions one would consistently form numerous embryoids while the other would form shoots.

Somaclonal variation is a general phenomenon of all plant regeneration systems that involve a callus phase viz., cultured explants, anthers, ovaries and microspores. Somaclonal variation was first reported for vegetative crop plants viz., potato, sugarcane but now this phenomenon is reported for cereals, legumes and other economically important plants (Table 4.1).

Chromosomal Variation

It has been known for sometime that chromosomal abnormalities occur at a high frequency in tissue cultures. In potato (tetraploid with 48 chromosomes) if plants are regenerated from tuber, leaf pieces or stem, nearly 10%

of such plants become aneuploid. Potato plants regenerated from protoplasts the extent of aneuploidy is much greater. It is well documented in sugarcane that chromosome number shifts can occur in cultures and culture derived plants. From a cytological study of sugarcane somaclonal variants, Liu and Chen (1976) concluded that there was no apparent correlation between morphological modifications and changes in chromosome number.

Table 4.1 Some examples of somaclonal variation in economically important species.

Species	Explant	Variant characters
Solanum tuberosum	Protoplast, leaf, callus	Tuber shape, yield, leaf maturity date, plant habit, stem, leaf and flower morphology, early and late blight resistance
Lycopersicon esculentum	Leaf	Male sterility, fruit colour, indeterminate growth
Saccharum officiarum	*Various*	*Eye spot, Fiji virus,* downy mildew, culmicolous smut disease, sugar yield, esterase isozyme
Triticum aestivum	Immature embryo	Plant height, spike *shape,* awns, maturity, tillering, α-amylase
Zea mays	Immature embryo	Endosperm and seedling mutants, *D. maydis* race, T toxin resistance, mt-DNA sequence rearrangement
Brassica species	Anthers, embryos, Meristems	Flowering time, growth habit, *Phoma lingam* tolerance

Applications of Somaclonal Variation

Those species which are amenable to cell culture, somaclonal variation may provide an important source of variability for plant improvement. The value of somaclonal variation to plant improvement depends in the ease with which given plant species can be cultured and plants regenerated from cell lines. Somaclonal variation may find its greatest application for plant improvement in concert with selection for desirable mutations at the cellular

level. It should no longer be surprising that in the recovery of cell culture mutants the frequencies have been high and mutagenic treatments often failed to noticeably enhance the frequency. Celluler selection is conceivable for the recovery of variants resistant to antimetabolites such as amino acid analogues, antibiotic drugs, pathotoxins, herbicides and physiological stress. Somaclonal variation, cellular selection and early rapid screening of regenerants collectively provide a powerful option for plant improvement.

19